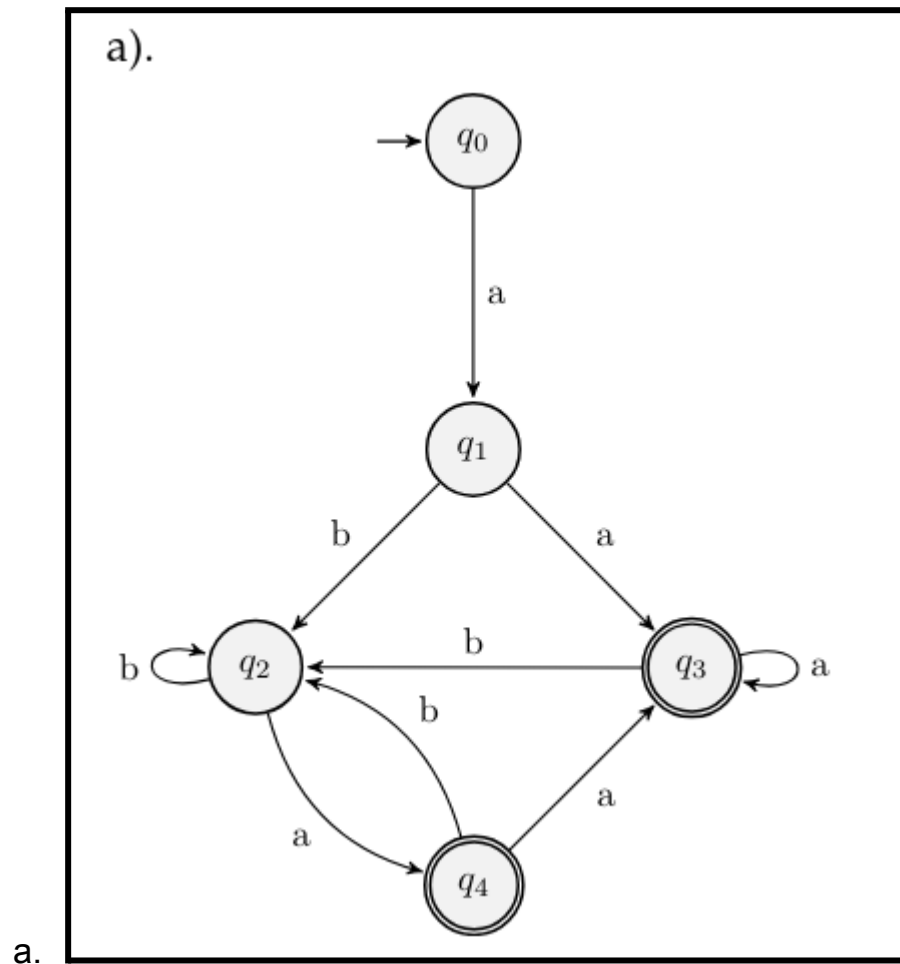


Aluno: Erick Molina Gehring
RA: 2251680

1. Determine o Autômato Finito Mínimo para cada um dos autómatos. Não esqueça de verificar os pré-requisitos necessários para o algoritmo de Minimização.



R:

a) ~~problema~~ q_0 , final: q_3, q_4

$F = \{q_3, q_4\}$
 $N = \{q_0, q_1, q_2\}$ q_d
 ↳ trap

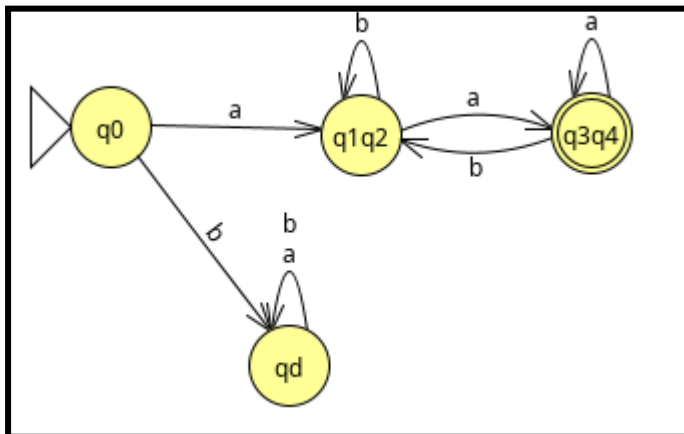
Setas equivalência

$\{q_3, q_4\}$	$\{q_1, q_2\}$
$q_3 \xrightarrow{a} q_3 (F)$ $q_3 \xrightarrow{b} q_2 (N)$ $q_4 \xrightarrow{a} q_3 (F)$ $q_4 \xrightarrow{b} q_2 (N)$	$q_1 \xrightarrow{a} q_3 (F)$ $q_1 \xrightarrow{b} q_2 (N)$ $q_2 \xrightarrow{a} q_1 (F)$ $q_2 \xrightarrow{b} q_2 (N)$
$\{q_0, q_d\}$ $q_0 \xrightarrow{a} q_1 (N)$ $q_0 \xrightarrow{b} q_d (N)$ $q_d \xrightarrow{a} q_d (N)$ $q_d \xrightarrow{b} q_d (N)$	

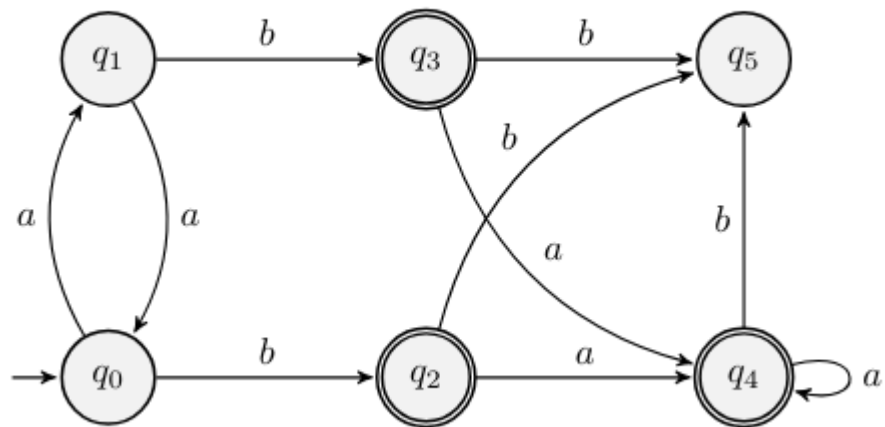
Para a transição

Equivalentes

Diferentes, pois $q_0 \xrightarrow{a} q_1$
 então ficam separados



b).



b.

R:

b) $F = \{q_2, q_3, q_4\}$
 $N = \{q_0, q_1, q_5\}$ qd

$\{q_0, q_1\}$
 $q_0 \xrightarrow{a} q_1 (F)$
 $q_1 \xrightarrow{a} q_0 (N)$
 $q_0 \xrightarrow{b} q_2 (F)$
 $q_1 \xrightarrow{b} q_3 (N)$

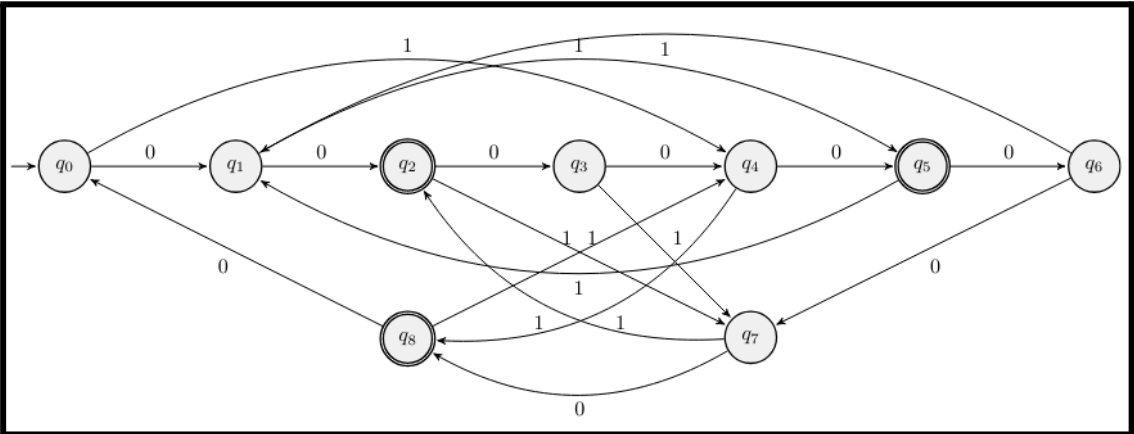
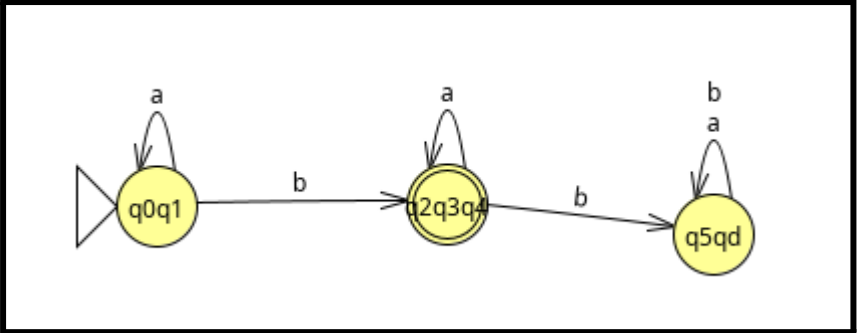
$\{q_2, q_3\}$
 $q_2 \xrightarrow{a} q_4 (F)$
 $q_3 \xrightarrow{a} q_4 (F)$
 $q_2 \xrightarrow{b} q_3 (N)$
 $q_3 \xrightarrow{b} q_5 (N)$

$\{q_0, q_1, q_2\}$
 $q_0 \xrightarrow{a} q_1 (F)$
 $q_1 \xrightarrow{a} q_0 (N)$
 $q_0 \xrightarrow{b} q_2 (F)$
 $q_1 \xrightarrow{b} q_3 (F)$
 $q_2 \xrightarrow{a} q_4 (F)$
 $q_2 \xrightarrow{b} q_3 (N)$

$\{q_0, q_1, q_2, q_3\}$
 $q_0 \xrightarrow{a} q_1 (F)$
 $q_1 \xrightarrow{a} q_0 (N)$
 $q_0 \xrightarrow{b} q_2 (F)$
 $q_1 \xrightarrow{b} q_3 (F)$
 $q_2 \xrightarrow{a} q_4 (F)$
 $q_2 \xrightarrow{b} q_3 (N)$
 $q_3 \xrightarrow{a} q_4 (F)$
 $q_3 \xrightarrow{b} q_5 (N)$

$\{q_0, q_1, q_2, q_3, q_4\}$
 $q_0 \xrightarrow{a} q_1 (F)$
 $q_1 \xrightarrow{a} q_0 (N)$
 $q_0 \xrightarrow{b} q_2 (F)$
 $q_1 \xrightarrow{b} q_3 (F)$
 $q_2 \xrightarrow{a} q_4 (F)$
 $q_2 \xrightarrow{b} q_3 (N)$
 $q_3 \xrightarrow{a} q_4 (F)$
 $q_3 \xrightarrow{b} q_5 (N)$
 $q_4 \xrightarrow{a} q_4 (F)$
 $q_4 \xrightarrow{b} q_5 (N)$

$\rightarrow (q_0, q_1) \xrightarrow{a} (q_2, q_3, q_4) \xrightarrow{a, b} (q_5, q_d)$



c.

R:

c) $F = \{q_0, q_5, q_8\}$

$N = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8\}$

$\{q_2, q_5\}$

$q_2 \xrightarrow{0} q_3 (N)$

$q_2 \xrightarrow{1} q_7 (N)$

$q_5 \xrightarrow{0} q_6 (N)$

$q_5 \xrightarrow{1} q_1 (N)$

Equivalente

$q_0 \xrightarrow{0} q_1$

$q_0 \xrightarrow{1} q_4$

$q_1 \xrightarrow{0} q_2$

$q_1 \xrightarrow{1} q_5$

$q_2 \xrightarrow{0} q_3$

$q_2 \xrightarrow{1} q_7$

$q_3 \xrightarrow{0} q_4$

$q_3 \xrightarrow{1} q_7$

$q_4 \xrightarrow{0} q_5$

$q_4 \xrightarrow{1} q_8$

$q_5 \xrightarrow{0} q_6$

$q_5 \xrightarrow{1} q_1$

$q_6 \xrightarrow{0} q_7$

$q_6 \xrightarrow{1} q_1$

$q_7 \xrightarrow{0} q_8$

$q_7 \xrightarrow{1} q_2$

$q_8 \xrightarrow{0} q_0$

$q_8 \xrightarrow{1} q_1$

$\{q_2, q_8\}$

$q_2 (N)$

$q_8 (N)$

$q_2 \xrightarrow{0} q_0 (N)$

$q_2 \xrightarrow{1} q_4 (N)$

Equivalente

$\{q_2, q_5, q_8\}$

$\{q_0, q_1\}$

$q_0 \xrightarrow{0} q_1 (N)$

$q_0 \xrightarrow{1} q_4 (N)$

$q_1 \xrightarrow{0} q_2 (F)$

$q_1 \xrightarrow{1} q_5 (F)$

$\{q_3, q_4\}$

$q_3 \xrightarrow{0} q_4 (N)$

$q_3 \xrightarrow{1} q_7 (N)$

$q_4 \xrightarrow{0} q_5 (N)$

$q_4 \xrightarrow{1} q_8 (F)$

$\{q_6, q_7\}$

$q_6 \xrightarrow{0} q_7 (N)$

$q_6 \xrightarrow{1} q_1 (N)$

$q_7 \xrightarrow{0} q_8 (F)$

$q_7 \xrightarrow{1} q_2 (F)$

$\{q_1, q_4, q_7\}$ equivalentes

$\{q_0, q_3, q_6\}$ equivalentes

